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ORIGINAL ARTICLE

Inter-individual variability in the adaptation of human muscle specific tension to progressive resistance training

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Abstract Considerable variation exists between people in the muscle response to resistance training, but there are numerous ways muscle might adapt to overload that might explain this variable response. Therefore, the aim of this study was to quantify the range of responses concerning the training-induced change in maximum voluntary contraction (MVC) knee joint torque, quadriceps femoris (QF) maximum muscle force (F), physiological cross-sectional area (PCSA) and specific tension (F/PCSA). It was hypothesized that the variable change in QF specific tension between individuals would be less than that of MVC. Fifty-three untrained young men performed progressive leg-extension training three times a week for 9 weeks. F was determined from MVC torque, voluntary muscle activation level, antagonist muscle co-activation and patellar tendon moment arm. QF specific tension was established by dividing F by QF PCSA, which was calculated from the ratio of QF muscle volume to muscle fascicle length. MVC torque increased by $26 \pm 11\%$ ($P < 0.0001$; range -1 to 52%), while F increased by $22 \pm 11\%$ ($P < 0.0001$; range -1 to 44%). PCSA increased by $6 \pm 4\%$ ($P < 0.001$; range -3 to 18%) and specific tension increased by $17 \pm 11\%$ ($P < 0.0001$; range -5 to 39%). In conclusion, training-induced changes in F and PCSA varied substantially between individuals, giving rise to greater inter-individual variability in the specific tension response compared to that of MVC. Furthermore, it appears that the change in specific tension is responsible for the variable change in MVC.

Keywords Quadriceps femoris · PCSA · Specific tension · Strength training · Variable training response

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Introduction

It is well established that resistance training induces gains in both strength and muscle size in humans. However, even in relatively small sample sizes, it is evident that inter-individual variation exists in these changes (Jones and Rutherford 1987; Narici et al. 1996b; Seynnes et al. 2009). Most previous studies have focused on the mean muscle response to overload and the variability was assumed to be experimental error associated with the measurements made or variation in the commitment of the subjects. More recently, however, there has been an increasing awareness that the variable training response is a "real" phenomenon and understanding the reasons for it may give insights into the mechanisms controlling muscle growth and changes in strength. In addition, there could be practical benefits in knowing who might or might not respond since training is an important aspect of sport and rehabilitation. Consequently, there is considerable interest in determining whether variants of particular genes are associated with the way in which human muscle adapts to strength training

(Bray et al. 2009). However, for a genetic study to provide useful information it is necessary to first characterize the variability in response, since there are several ways in which a muscle may increase in strength. Broadly, a muscle may increase in quantity [defined as the physiological cross-sectional area (PCSA)] or quality (characterized as specific tension; force per unit PCSA), and there may also be questions concerning muscle activation and the co-contraction of agonist and antagonist muscles, as well as muscle architecture.

A large-scale investigation has reported substantial variability in the response to 12 weeks of unilateral resistance training of the elbow flexor muscle group in 243 young men (Hubal et al. 2005). Relative training-induced changes in strength varied between individuals from 0 to 150% and from −24 to 149%, according to the maximum load lifted in one repetition (1 RM) and the maximum isometric voluntary contraction (MVC), respectively. Relative changes in muscle anatomical CSA, on the other hand, ranged from −3 to 56%. Thus, not only does the response to strength training vary considerably between individuals but also the amount of variation appears to differ, according to the variable measured. Although a proportion of the gain in 1 RM will undoubtedly be due to an increase in muscle force, an unknown amount is likely to result from increased co-ordination of additional muscles during training (Rutherford and Jones 1986), and the acquisition of this skill might differ between people. Furthermore, an MVC does not necessarily represent the maximum isometric force produced by a fully activated muscle. To assess this in vivo, the ability of an individual to fully activate their muscle, as well as the level of antagonist muscle co-activation and the length of the moment arm are critical measurements (Erskine et al. 2009; Maganaris et al. 2001; Reeves et al. 2004). Moreover, as most human skeletal muscles are pennate in that the muscle fibres are aligned at an angle to the line of pull of the tendon (Alexander and Vernon 1975), the muscle PCSA is the most appropriate measure to provide an accurate assessment of training-induced muscle hypertrophy, as this parameter represents the total CSA of all the constituting fibres of a muscle at right-angles to their long axes, and is the main determinant of maximum muscle force (Powell et al. 1984). Therefore, an accurate quantification of the full range of responses of human muscle to resistance training requires the force and PCSA of a maximally activated muscle or group of muscles to be determined, thus enabling the extent of variable change in muscle specific tension to be established.

The aim of the present study was to quantify the range of human muscle responses to resistance training with regard to changes in quadriceps femoris (QF) muscle strength and size. By accounting for training-induced alterations in voluntary activation, antagonist muscle co-activation, muscle volume and architecture, the main hypothesis of the study was that the inter-individual variation in the change in QF muscle specific tension would be less than that in MVC knee joint torque.

Materials and methods

Participants

Potential participants were excluded if they had history of lower-limb fracture, their age was outside the range of 18–39 years, had taken part in strength training within the 12 months prior to the study, had used dietary supplements or performance enhancing aids or if they were considered to be in ill health (assessed from the responses to a general health questionnaire). The study was approved by the Manchester Metropolitan University local ethics committee. Fifty-three untrained males, aged 20 ± 3 years, height 178 ± 5 cm, body mass 77 ± 11 kg (mean ± SD), were eligible to participate in the study after written informed consent was obtained. Volunteers were familiarized with all test procedures and equipment within a 14-day period prior to the baseline measurements and, as far as possible, testing and training sessions occurred at the same time of day for each participant.

Habitual physical activity level

The habitual physical activity level (PAL) of each participant was assessed by questionnaire (Baecke et al. 1982) prior to commencing the 9-week training period. The overall PAL was scored using a scale from 1 to 5 points, where 1 was the least active, 3 was intermediate and 5 was extremely active. Participants were asked to maintain their PAL and habitual dietary intake over the course of the study.

Experimental design

Maximum QF muscle force, PCSA and specific tension were determined in previously untrained individuals, as previously described in detail in Method 2 of Erskine et al. (2009), before and after 9 weeks of leg-extension resistance training.

Resistance training

The strength training protocol has been reported in detail elsewhere (Erskine et al. 2010). Briefly, unilateral knee extension exercises were performed three times a week for a period of 9 weeks on the right leg. The single repetition maximum (1 RM) was determined at the beginning of the

training programme and reassessed at the start of each week on a standard leg-extension machine (Technogym SpA, Gambettola, Italy). The training intensity was set relative to the 1 RM and the training load was therefore increased progressively throughout the training period. Each training session consisted of a warm-up set of 10 repetitions at 40% of the 1 RM, followed by four sets of 10 repetitions at 80% of the 1 RM, with a 2-min rest period between each set. Each repetition required approximately 1-s to lift and 1-s to lower the load through a range of 110°–20° of knee flexion (0° = full leg extension). All training sessions were supervised and appropriate verbal encouragement was given throughout each session.

Maximum isometric force

Maximum voluntary contraction

Isometric leg-extension MVC torque was assessed using an isokinetic dynamometer (Cybex Norm, Cybex International, NY, USA), with the knee joint angle set at 70°–90° knee flexion (depending on the angle at which peak torque was produced, defined as the optimum angle). Each MVC lasted 2–3 s and there was a rest-interval of 60 s between each contraction. During this protocol, the participant was firmly strapped at the hip, distal thigh and chest with inextensible straps to minimize movement.

Antagonist muscle co-activation

To determine the extent of antagonist muscle co-activation during the leg extension MVC, the electromyographic (EMG) activity of the biceps femoris muscle was recorded using surface electrodes (Neuroline, Medicotest, Rugmarken, Denmark). The precise location of the electrodes was marked on the participant's skin and recorded on an acetate sheet to ensure accurate electrode repositioning in subsequent trials. The root mean square of the raw EMG signal was calculated over 1 s around the peak torque during each MVC at optimum joint angle. Biceps femoris muscle EMG activity was recorded during isometric knee flexion MVC at optimum joint angle (Kellis and Baltzopoulos 1997), and the torque produced by the hamstrings during knee extension MVC was estimated assuming a linear relationship between torque and EMG activity (Reeves et al. 2004). The addition of the estimated antagonist torque during MVC knee extension to the actual MVC knee extension torque provided the overall MVC knee extension torque.

Voluntary muscle activation

The interpolated twitch technique (Rutherford et al. 1986b) was used to determine voluntary QF muscle activation, which was assessed with the participant seated in the strength-testing chair as explained above and the knee joint fixed at 90° flexion. Electrical stimulation was administered via two 7.5 cm × 12.5 cm self-adhesive electrodes (Versastim, Conmed, NY, USA) placed distally and proximally over the QF. The QF was stimulated in a relaxed state with 100 mA pulses of 200 μs which were increased in 50 or 100 mA increments (Digitimer stimulator, model DS7, Welwyn Garden City, UK) until no further increase in torque was observed. This current was used 2 min later to elicit a single twitch whilst the participant was in the resting state, followed 5-s later by a second twitch during MVC. True maximal torque (TMT) was calculated as:

$$\mathrm{TMT} = \mathrm{MVC}_{\mathrm{CORR}}/(1 - t/T)$$

where t is the amplitude of the superimposed twitch, T the value of the twitch before the MVC and $\mathrm{MVC}_{\mathrm{CORR}}$ is the MVC corrected for co-activation. The percentage activation was given by

$$100(1 - t/T)$$

The level of activation at 90° knee flexion was applied to the $\mathrm{MVC}_{\mathrm{CORR}}$ torque value at the optimum joint angle to determine TMT.

Patellar tendon moment arm

The patellar tendon moment arm (d_{PT}) was quantified using a 0.2-T magnetic resonance imaging (MRI) scanner (G-Scan, Esaote Biomedica, Genoa, Italy). Sagittal and coronal-plane knee scans were acquired using a Turbo 3D T1-weighted sequence with the following scanning parameters: time of repetition (TR) 40 ms; time to echo (TE) 16 ms; matrix 256 × 256; field of view 180 mm × 180 mm; slice thickness 3.4 mm; interslice gap 0 mm. The knee was scanned at rest with the participant in the supine position and the knee joint fully extended. Images in the coronal plane were used to identify the appropriate sagittal scans, which were used to locate the tibiofemoral contact point (TFCP). d_{PT} was defined as the length of the perpendicular line between the axis of the patellar tendon and the TFCP (Tsaopoulos et al. 2006). Previously reported values of d_{PT} length at full extension (0° knee flexion) and optimum angle (Baltzopoulos 1995) were used to estimate d_{PT} at optimum knee joint angle. This value was subsequently multiplied by 1.14 to account for a 14% increase in d_{PT} from rest to MVC (Tsaopoulos et al. 2007).

Maximum isometric QF force (F) was thus calculated as:

$$F = \mathrm{TMT}/d_{\mathrm{PT}}$$

The test–retest reliability for calculating F using the above method has been reported previously (Erskine et al. 2009) in the form of the coefficient of variation (CV),

3.99%, and intraclass correlation coefficient (ICC), 0.92 (0.77–0.98).

Physiological cross-sectional area

Muscle volume

Femur length was measured as the distance from the tibiofemoral point of contact to the proximal insertion of the VL muscle, identified using ultrasonography (MyLab25, Esaote, Genova, Italy). Oil-filled beads were attached to the skin at the measurement location (corresponding to 40% of femur length from the distal end). The participant was scanned in the supine position, and five serial axial slices were acquired around the position of the beads using a spin echo T1-weighted protocol with the following scanning parameters: TR 200 ms; TE 1.9 s; matrix 256 × 256; field of view 180 mm × 180 mm; slice thickness 5 mm; interslice gap 5 mm. Together with the femur length, the ACSA of each constituent QF muscle and a series of regression equations, QF volume was estimated by adapting a previously published method (Morse et al. 2007).

Muscle architecture

During MVC at optimum knee joint angle, ultrasound scans of the vastus lateralis (VL) muscle fascicle length (L_f) and pennation angle (θ_p) were acquired at 50% of the muscle length along the mid-sagittal plane and L_f and θ_p were subsequently measured using digitizing software (NIH ImageJ, version 1.39b, National Institutes of Health, Bethesda, USA). Consequently, QF PCSA was calculated by dividing the estimated QF volume by VL L_f. The CV and ICC for calculating QF PCSA using the above technique are 5.0% and 0.90 (0.70–0.97), respectively (Erskine et al. 2009).

Specific tension

As the force generated by the QF muscle is reduced according to the θ_p when resolved along the tendon, QF PCSA was multiplied by the cosine of the VL θ_p [which can be used as an approximation of the θ_p for the whole QF muscle (Erskine et al. 2009)] to provide the reduced QF PCSA. Force was subsequently divided by the reduced QF PCSA to provide QF specific tension. The CV and ICC for this measurement are 9.1% and 0.69 (0.26–0.89), respectively (Erskine et al. 2009).

Statistics

Post-training differences in MVC, co-activation, voluntary activation, maximum force, muscle volume, muscle fascicle length, muscle fascicle pennation angle, muscle PCSA and specific tension were analysed using paired Student's *t* tests. A Pearson correlation was used to determine the relation between the changes in maximum force and the changes in PCSA. Multiple regression analysis determined the relationships between the training-induced change in MVC torque and changes in voluntary activation, co-activation, muscle volume, muscle architecture, PCSA and specific tension. Statistical significance was accepted when $P < 0.05$. All data are presented as mean ± standard deviation (SD) unless otherwise stated. The variable change in MVC, maximum force, PCSA and specific tension was reported for each parameter as the single sample CV [100 (SD mean^{-1})]. Individuals were defined as high or low responders to resistance training if their relative post-training changes were, respectively, ≥1 SD or ≤1 SD.

Results

Habitual physical activity level

The PAL of the 53 participants prior to starting the strength-training period was 2.7 ± 0.3 points, described as slightly less than "intermediate" in terms of PAL scores (Baecke et al. 1982). PAL scores did not correlate with any of the muscle changes recorded after training.

Single repetition maximum

Following 9 weeks of high-intensity unilateral leg-extension exercise, the 1 RM increased by 68 ± 30% ($P < 0.001$; Table 1), with individual changes ranging from 18 to 113%.

Maximum isometric QF force

Maximum voluntary contraction (MVC) torque increased by 26 ± 11% ($P < 0.001$; Table 1) and the full range of individual responses is shown in Fig. 1; the CV for the change in MVC was 42%. Eight participants were designated high responders (≥1 SD) and 8 were classed as low responders (≤1 SD). QF voluntary activation increased by 3.8 ± 5.2% ($P < 0.001$; range −5 to 18%; Table 1), while antagonist muscle co-activation did not change significantly with training (range −13 to 15%; Table 1). Similarly, as optimum knee joint angle did not change, no alteration in d_{PT} was observed with training (range −4.8 to 4.4%; Table 1). Consequently, maximum isometric QF force increased by 22 ± 11% ($P < 0.001$; Table 1). The complete account of individual responses is shown in Fig. 2 (CV = 51%), from which it can be seen that 11 of the participants were high responders (of which only 5

Table 1 Measured and calculated variables before and after training; $n = 53$; mean ± SD

Variable	Before training	Post training
1 RM (kg)	54.3 ± 10.9	88.2 ± 12.0*
MVC (Nm)	246 ± 51	310 ± 62*
Co-activation (%)	13.9 ± 4.9	13.6 ± 4.9
Activation (%)	92.8 ± 5.2	96.2 ± 3.8*
d_{PT} length (cm)	4.8 ± 0.3	4.8 ± 0.3
F (N)	5,722 ± 1,036	6,968 ± 1,343*
V_m (cm^3)	2,080 ± 254	2,201 ± 277*
L_f (cm)	8.9 ± 1.3	8.9 ± 1.4
θ_p (°)	16.1 ± 2.3	18.2 ± 2.9*
PCSA (cm^2)	237 ± 41	251 ± 43*
Specific tension (N cm^{-2})	25.6 ± 5.1	29.8 ± 6.4*

MVC maximal voluntary contraction torque, d_{PT} patellar tendon moment arm, F maximum muscle force, V_m quadriceps femoris muscle volume, L_f fascicle length, θ_p fascicle pennation angle, *PCSA* quadriceps femoris muscle physiological cross-sectional area

* $P < 0.001$ significantly different from baseline

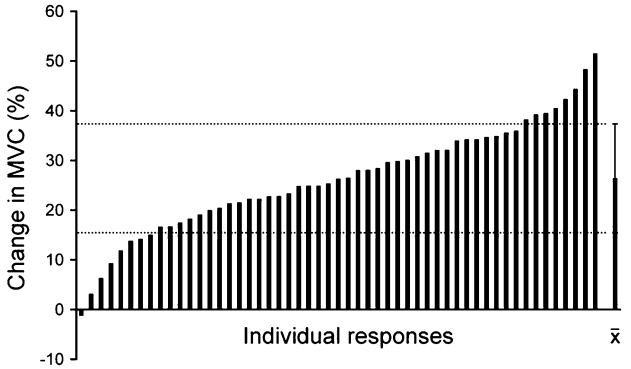


Fig. 1 The range of individual changes in MVC relative to baseline values; *dotted lines* represent ±1 SD from the mean ($\bar{x}$), from which high responders (≥1 SD) and low responders (≤1 SD) were determined; $n = 53$

were also high responders in relation to changes in MVC) and 10 were low responders (of which only 6 were also low responders concerning changes in MVC).

Physiological cross-sectional area and specific tension

Following training, QF muscle volume increased by 5.6 ± 3.4% ($P < 0.001$; range 0–16%; Table 1). Since VL muscle L_f did not change (range −9 to 13%; Table 1), QF PCSA increased by 5.7 ± 4.5% ($P < 0.001$; Table 1) and the full range of responses is shown in Fig. 3 (CV = 75%), from which 8 individuals were defined as high responders and 8 were low responders. VL muscle θ_p increased by 13 ± 10% ($P < 0.001$; range −5 to 39%; Table 1) and when all these factors were taken into account, QF muscle specific tension increased by 16.5 ± 11.2% after training

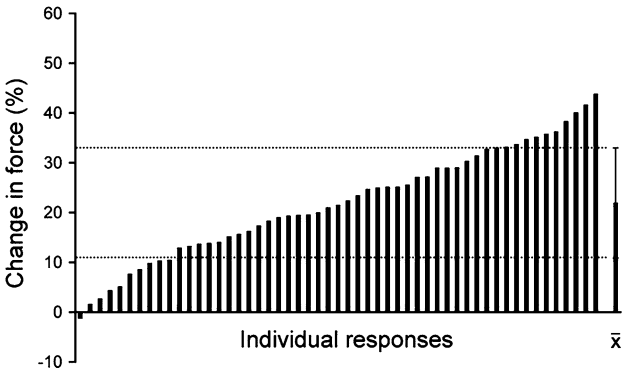


Fig. 2 The range of individual changes in maximum muscle force relative to baseline values; *dotted lines* represent ±1 SD from the mean ($\bar{x}$), from which high responders (≥1 SD) and low responders (≤1 SD) were determined; $n = 53$

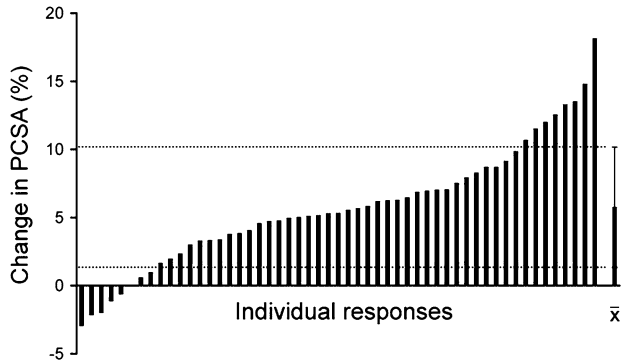


Fig. 3 The range of individual changes in muscle PCSA relative to baseline values; *dotted lines* represent ±1 SD from the mean ($\bar{x}$), from which high responders (≥1SD) and low responders (≤1 SD) were determined; $n = 53$

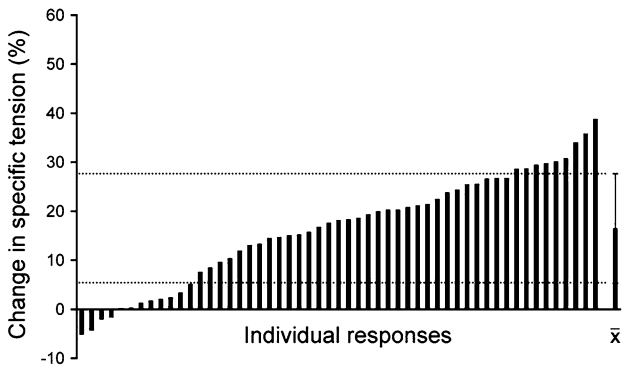


Fig. 4 The range of individual changes in muscle specific tension relative to baseline values; *dotted lines* represent ±1 SD from the mean ($\bar{x}$), from which high responders (≥1 SD) and low responders (≤1 SD) were determined; $n = 53$

($P < 0.001$; Table 1). The CV for the change in specific tension was 68% and all the individual changes are shown in Fig. 4, including the high ($n = 9$) and low ($n = 12$) responders. None of the PCSA low responders were

classified as low responders according to change in force or specific tension, although interestingly, one was classed as a high responder regarding change in muscle force. However, 75% of the low specific tension responders were also classified as low responders according to changes in muscle force. Regarding specific tension high responders, 89% were also high muscle force responders, of which only one was classified as a high responder regarding a change in PCSA. The discrepancy between high and low PCSA and specific tension response groups is supported by the lack of correlation between the training-induced changes in PCSA and maximum force ($r = 0.14$; $P = 0.29$). Furthermore, multiple regression analysis revealed the following relationships between the training-induced changes in MVC knee joint torque and changes in voluntary activation ($r = 0.47$; $P = 0.06$), antagonist muscle co-activation ($r = 0.09$; $P = 0.475$), QF muscle volume ($r = 0.15$; $P = 0.439$), fascicle length ($r = -0.47$; $P = 0.058$), fascicle pennation angle ($r = -0.33$; $P = 0.213$), QF muscle PCSA ($r = 0.48$; $P = 0.047$) and QF muscle specific tension ($r = 0.79$; $P < 0.001$).

Discussion

The present study is the first to quantify the complete range of responses of human muscle to resistance training, regarding changes in maximum force, PCSA and specific tension. The main conclusion is that the major variation in strength as a result of training appears to be due to large inter-individual variability in the training-induced adaptation of muscle specific tension.

Since it has been suggested that prior training status affects the response to strength training (Kraemer et al. 2002), it might be assumed that the variable response to strength training observed in the present study was influenced by different levels of habitual physical activity (PAL) or training status prior to the commencement of training. However, prior to taking part in the study all participants were screened for previous experience of strength training and their PAL score was found to be less than intermediate. Moreover, training-induced changes in knee joint torque and muscle specific tension did not correlate with either baseline levels or PAL. Therefore, it is maintained that the response spectrums observed in the present study are real reflections of how the QF muscle adapts to the particular type and intensity of resistance training performed in the current investigation.

The present study observed a large variable change (CV = 45%) in the unilateral leg extension single repetition maximum (1 RM), which was similar to the previously reported variability (CV = 55%) in the unilateral preacher curl 1 RM in young men (Hubal et al. 2005). Numerous resistance training studies in humans have used the 1 RM as a method of determining the effect of resistance training on muscle strength (Charbonneau et al. 2008; Hand et al. 2007; Hubal et al. 2005; Kostek et al. 2005; Riechman et al. 2004). However, although the 1 RM is a relatively simple task to perform, it involves not just the intrinsic strength of the muscle group but also the actions of synergistic and stabilizing muscles (Rutherford and Jones 1986), thus questioning its suitability for detailed physiological assessment. In contrast, maximum voluntary isometric contraction (MVC) provides a more accurate assessment of muscle strength, and the 26% increase in MVC knee extension torque observed in the present study is comparable to that found in previous investigations into the effects of training on the knee extensors (Jones and Rutherford 1987; Narici et al. 1989; Rutherford et al. 1986a) and other muscle groups (Cureton et al. 1988; Hubal et al. 2005; Kawakami et al. 1995). Great care was taken in the present study to ensure that the subjects were similar in their body size, age and habitual physical activity before training, that participant compliance was good and the training supervised by the same investigators throughout. Nevertheless, the range of training responses for MVC torque seen (CV = 42%) was large but somewhat less than that reported previously for the upper limb (CV = 107%) (Hubal et al. 2005). However, changes in MVC do not necessarily reflect changes in the contractile properties of the muscle itself, and for this it is necessary to determine the true muscle force, taking into account changes in voluntary muscle activation, antagonist muscle co-activation and moment arm length (Erskine et al. 2009; Maganaris et al. 2001; Reeves et al. 2004). Yet, rather than reducing the variation, taking these factors into account increased the variability of the training response (CV of 51% for muscle force vs. 42% for MVC torque).

As muscle force is primarily determined by the number of sarcomeres arranged in parallel, which is represented by the PCSA (Close 1972; Powell et al. 1984), it is important to know how resistance training affects PCSA in order to ascertain a complete overview of the variable response of human muscle to overload. Previous studies have used whole muscle volume (Charbonneau et al. 2008; Kanehisa et al. 2002) or ACSA (Ferri et al. 2003; Hubal et al. 2005; Tracy et al. 1999) to determine the effect of resistance training on muscle size. However, unlike PCSA, neither of these two indices of muscle size represents the CSA of the total number of fibres at right angles to their axes in pennate muscles. Although the mean increase in PCSA in the present study was relatively small (6%), it was evident that this change varied greatly between people (CV = 75%). Of the eight low hypertrophic responders in the present study, five exhibited a slight (0 to −3%) decrease in PCSA, which was due to muscle fascicle length increasing

relatively more than muscle volume, thus suggesting that the number of serial sarcomeres increased at the expense of parallel-aligned sarcomeres. However, as the negative changes are smaller than the test–retest coefficient of variation for this measurement (5%), it is possible that these changes are due to the small measurement error and there was in fact no change in PCSA in these five individuals. In 89% of the participants, however, a varying degree of hypertrophy did occur and eight were classified as high responders. In each of these high responders, muscle volume increased relatively more than fascicle length, implying an increase in CSA of the composing fibres. This highlights the different ways in which human muscle adapts to overload training. Of the eight high responders, only two were also classified as high responders regarding a change in maximum force. Moreover, one of the low hypertrophic responders was in the high response group for change in maximum force. Furthermore, the change in muscle force did not correlate with the variable change in PCSA and it is evident therefore that while there is a variable hypertrophic response, this does not explain the variability in the force response to training. These unexpected and rather astonishing findings cannot be explained by the results from the present study but they are most probably due to inter-individual differences in the way muscle adapts to resistance training. For example, some people may have increased their strength through an increase in contractile material and therefore a proportional increase in muscle PCSA, while others may have increased the ability to transmit force laterally, leading to larger increases in force than PCSA (discussed below).

It is acknowledged that certain assumptions were made in the calculation of in vivo muscle PCSA, which might have affected the estimation of specific tension and also the lack of relationship between the changes in muscle force and PCSA after training. Muscle PCSA was calculated by dividing muscle volume by fascicle length (Alexander and Vernon 1975; Haxton 1944; Maganaris et al. 2001; Reeves et al. 2004), with the latter measured during MVC in the mid-sagittal plane of the central portion of the muscle. Therefore, it was assumed that the muscle was cylindrical in shape and that its fascicles were a constant length. Previous in vivo (Maganaris and Baltzopoulos 1999; Maganaris et al. 1998; Narici et al. 1996a) and cadaveric (Friederich and Brand 1990) findings in human muscles support the assumption that fascicle length is relatively constant along the length of a muscle. Furthermore, this calculation of muscle PCSA does not account for a possible change in intramuscular fat content, myofibrillar packing or in muscle fibre-type composition. The first two possibilities could feasibly lead to an under or overestimation of muscle PCSA, which would lead to erroneous calculations of muscle specific tension. However, it has been shown that the level of non-contractile material in the muscle of young, healthy male individuals is relatively small (Holmback et al. 2002), while previous studies have demonstrated that the specific tension of isolated single muscle fibres does not change following 12 weeks of resistance training (Trappe et al. 2001; Trappe et al. 2000; Widrick et al. 2002), thus suggesting that there is no change in myofibrillar packing following short-term strength training. Some investigations have provided evidence that single fibre specific tension is fibre-type dependent (Bottinelli et al. 1996; D'Antona et al. 2006; Pansarasa et al. 2009), thus leading to the possibility that a training-induced change in fibre-type composition might cause a change in the muscle force–size relationship. However, other investigators have demonstrated that there is no difference in isolated single fibre specific tension between fibre-types (Gilliver et al. 2009; Larsson and Moss 1993; Ottenheijm et al. 2005). Therefore, it remains unclear whether a change in fibre-type composition would have influenced the lack of correlation between training-induced changes in muscle force and PCSA.

The cause for the varied increase in muscle specific tension between people following strength training is not known and cannot be inferred from the results of the present study. It is acknowledged that a small amount of this variance might have been induced by an accumulation of measurement error introduced by the multitude of measurements necessary to calculate muscle specific tension in vivo. However, even if a more basic approach of dividing knee joint MVC by QF ACSA (two primary, highly reproducible measurements) had been adopted, the inter-individual change in MVC/ACSA (54%) would have been greater rather than smaller than that of MVC alone (42%), thus supporting the specific tension data of the present study. A change in fibre-type expression appears to be an unlikely explanation for the differences between people regarding the change in specific tension, since it has been shown in previous studies that strength training tends to decrease the expression of the fast IIX myosin isoform (Adams et al. 1993; Campos et al. 2002; Fry et al. 1994; Roman et al. 1993; Staron et al. 1994) and there appears to be no difference between the specific tension of type IIA and IIX fibres (Bottinelli et al. 1996; D'Antona et al. 2006; Pansarasa et al. 2009). Another possibility is that skeletal muscle in these individuals adapts by increasing its ability to transmit force laterally to the tendon. This increase might occur through an increase in attachments between intermediate sarcomeres and the extracellular matrix (Jones et al. 1989), either as a consequence of an increase in collagen synthesis (Miller et al. 2005) or proteins that constitute structural complexes known as costameres (Kosek and Bamman 2008; Woolstenhulme et al. 2006). The consequence of this is that the "effective" PCSA of

the muscle would increase, thus allowing muscle force to increase without additional contractile material being required although the effective length of the fibres would decrease and the power remain relatively constant (Jones et al. 1989). Furthermore, it remains to be seen whether the low responders of the present study may have expressed a different training response had an alternative training design been adopted. For example, if the participants had performed purely eccentric training rather than classic leg extension training that incorporates both concentric and eccentric modes of muscle contraction, a greater increase in muscle fascicle length (Reeves et al. 2009) might have been observed, which would have reduced the increase in PCSA and led to a greater increase in specific tension.

Conclusions

The results from the present study clearly demonstrate that the response of human muscle to high-intensity resistance training varies considerably between previously untrained young men. Large differences in the hypertrophic response of the muscle were not related to the variable response of muscle force, giving rise to variable increases in the specific tension of the muscle. Contrary to the main hypothesis of the study, the inter-individual variability in the muscle specific tension response to resistance training was larger than that of MVC joint torque. However, this greater variable response appears to be the main determinant of the variation in the MVC adaptation to training. In searching for mechanisms underlying these variable changes, whether of a genetic or other nature, future investigations should focus not only on factors involved in muscle growth but also on those that affect the quality, or force generating capacity, of skeletal muscle.

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